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Review

Feed frame: The last processing step before the tablet compaction in pharmaceutical manufacturing

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ABSTRACT

The feed frame is a force-feeding device used in the die filling process. The die filling process is crucial within pharmaceutical manufacturing to guarantee the critical quality attributes of the tablets. In recent years, interest in this unit has increased because it can affect the properties of the powder blend and tablets, and because of the success in real time monitoring of powder blend uniformity potential for Process Analytical Technology as described in this review. The review focuses on the recent advances in understanding the powder flow behavior inside the feed frame and how the residence time distribution of the powder within the feed frame is affected by the operating conditions and design parameters. Furthermore, this review also highlights the effect of the paddle wheel design and feed frame process parameters on the tablet weight, the principal variable for measuring die filling performance.

1. Introduction

The top 20 U.S. best-selling drugs in 2018, included sales from tablets and capsules of greater than $27 billion (Blankenship, 2019). Tablets are expected to remain the preferred dosage form in the worldwide market, with sales estimated to surpass $500 billion by the end of 2027 (Langhauser, 2019). In 2018, 53% of new molecular entities (NMEs) approved by the U.S. Food and Drug Administrationś (FDA) Center for Drug Evaluation and Research (CDER) were for solid oral dosage products, with 24 products approved as tablets and seven as capsules (FDA, 2019; Langhauser, 2019). Tablets offer many advantages, such as physical and chemical stability, dosage reliability, easy administration, simplicity of manufacture, and lower unit cost of production (Langhauser, 2019; Méndez et al., 2012a; Natoli et al., 2017). Regardless of the manufacturing route (i.e., direct compaction, wet granulation, dry granulation or hot melt extrusion) tablets are obtained in the tableting process where the powders are transformed into a dense compact. The tableting process is a continuous operation that involves three consecutive stages: die filling, compression, and decompression and ejection of the tablet. However, the performance of the tableting process depends directly on die filling (Goh et al., 2018a). The die filling process is critical to determine the amount of powder in the dies and therefore the drug content that a patient receives (Goh et al., 2018a; Van Snick et al., 2018).

Most tablets are produced using a rotary tablet press, where die filling is done using a forced feeder, called feed frame. Modern pharmaceutical rotary tablet presses can produce more than 1.5 million tablets per hour (Dühlmeyer et al., 2018). The feed frame is an essential unit within the tablet manufacturing process because it defines the amount of the powder to be compacted and can affect its composition through segregation or additional mixing; therefore, defining tablet weight and its drug content. A non-uniform die filling may significantly affect the compression force, and consequently, the physicochemical properties of the tablets, such as density and porosity (Armstrong, 2007; Natoli et al., 2017; Pawar et al., 2016; Van Snick et al., 2017), tensile strength and hardness (Méndez et al., 2012a; Pawar et al., 2016; Van Snick et al., 2018, 2017), disintegration and dissolution rate (Méndez et al., 2012a; Natoli et al., 2017; Van Snick et al., 2017) and weight variability and drug content (Dühlmeyer et al., 2019a; Goh et al., 2018a, 2018b; Grymonpré et al., 2018b, 2018a; Mehrotra et al., 2009; Méndez et al., 2012a; Peeters et al., 2015; Yaginuma et al., 2007).

In the feed frame, the powder blend is fed into moving dies on a rotating turret using paddle wheels. Different mechanisms act simultaneously in a feed frame: gravity feeding (where powder falls from the hopper to feed frame, and then into the dies), forced feeding (by rotating feed frame paddle wheels), suction fill (when the bottom punch drops rapidly, it creates a partial vacuum that drives the powder to the dies), centrifugal forces (by rotational movement of the die disc and the

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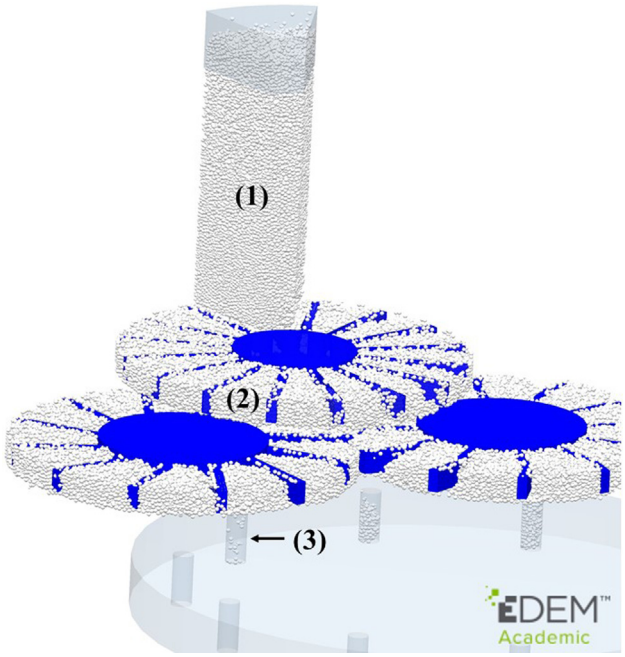


Fig. 1. Representation of the 3-stages of die filling by forced feeder (1) Powder flows through the hopper to feed frame (2) Powder flows through a confined space (3) Transfer of powder blend from the feed frame to the dies.

paddle wheels) and overhead pressure (due to the pressure exerted on the powder in the feeder and die by the weight of the powder) (Jackson et al., 2007; Peeters et al., 2015; Sinka et al., 2004; Van Snick et al., 2018). Fig. 1 illustrates the three main stages of the die filling process in a forced feeder. First, the powder flows through the hopper or chute to the feed frame, second, the powder flows to the dies within the feed frame promoted by paddle wheel movement through a confined space, and finally, the powder is transferred from the feed frame to the dies (Mateo-Ortiz and Méndez, 2015). This review focuses on the second stage, the powder flow in confined space since the first stage (Anand et al., 2008; Baxter et al., 2000; Hildebrandt et al., 2019a; Ketterhagen et al., 2008, 2009; Ketterhagen and Hancock, 2010; Liss et al., 2004; Prescott and Hossfeld, 1994) and the third stage (Coube et al., 2005; Goh et al., 2018a; Mehrotra et al., 2009; Mills and Sinka, 2013; Schneider et al., 2007; Sinka et al., 2004; Wu, 2008; Wu and Cocks, 2004; Zakhvatayeva et al., 2018) have been widely discussed by several groups.

The different designs of the feed frame depend on the tablet press model. The feed frame can contain up to three chambers with a paddle wheel to control the feeding system (Goh et al., 2018a). Fig. 2(a) shows a one-chamber feed frame. This feed frame is usually found in laboratory-scale tablet presses, such as the Korsch XL100 rotary tablet press (Korsch AG, Berlin, Germany). Fig. 2(b) illustrates a two-chamber feed frame, where the two chambers may or may not be in a coplanar position. For instance, the Manesty Betapress feed frame (Manesty Machines Ltd, Liverpool, England) has the chambers in two planes, while the Killian T-100 tablet press feed frame (IMA Kilian GmBH & Co. KG, Köln, Germany) has the two chambers in a coplanar position. An advantage of the two-chamber feed frame is that it has a die filling region for each chamber, which increases the time that the die remains within the die fill region, therefore, a greater die filling is promoted. Fig. 2(c) shows a three-chamber feed frame. This feed frame has two levels; in the top-level, the paddle wheel of the distributing chamber reduces the powder consolidation entering the feed frame. In the bottom level, the paddle wheels in the filling and dosing chamber are responsible of the die filling. In addition to having the two die filling regions, one of the advantages of three-chamber feed frame is the greater quantity of material available to feed the dies (Dühlmeyer et al., 2018). The three-

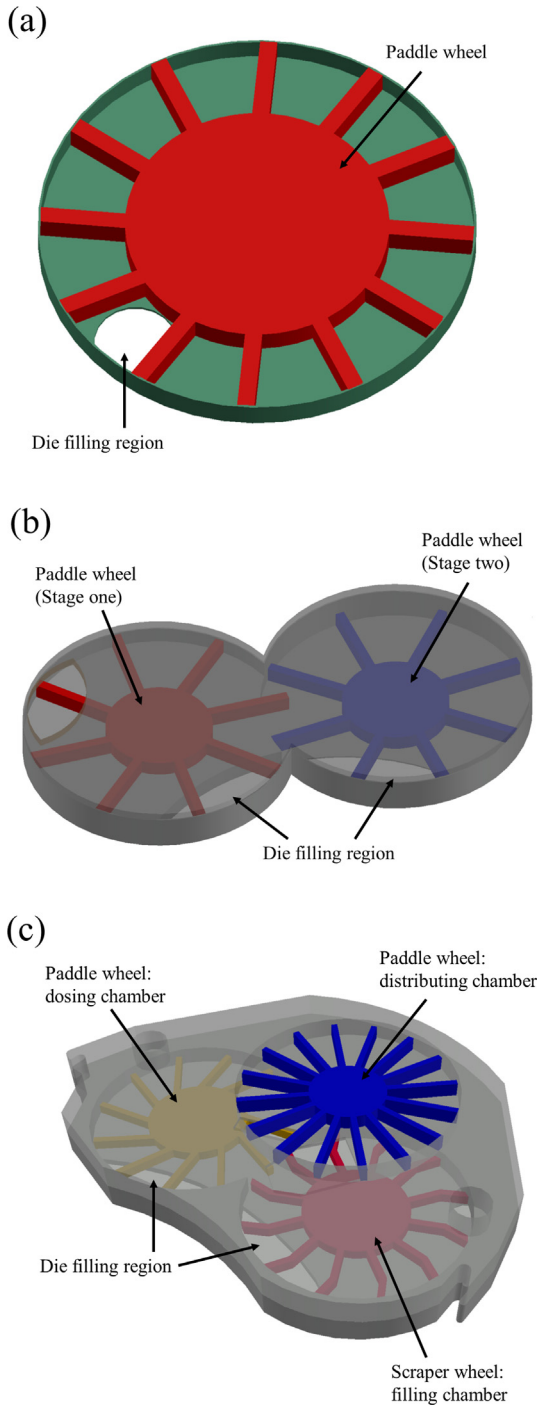


Fig. 2. Schematic representation of feed frames of (a) one, (b) two, and (c) three-chambers.

chamber feed frame can be found in industrial tablet presses, such as the Fette 3090, Fette 1200i and Fette 102i (Fette Compacting, Schwarzenbek, Germany). Different paddle wheel designs in terms of number of paddles, height, shape and geometry have been developed to improve the performance of the die filling process (De Leersnyder et al., 2018; Grymonpré et al., 2018b; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016; Méndez et al., 2010).

The feed frame is the last processing step that allows access to flowing powder before the tablet compaction. In the last few years, a significant number of articles related to the feed frames have been published, including discrete element method (DEM) simulations which have increased the understanding of powder flow behavior

(Hildebrandt et al., 2019a, 2019b, 2018; Ketterhagen, 2015; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016, 2015). These recent studies include several real-time spectroscopic methods developed for monitoring blend uniformity within the feed frame (Dalvi et al., 2019, 2018; De Leersnyder et al., 2018; Durão et al., 2017; Gosselin et al., 2017; Hetrick et al., 2017; Li et al., 2019, 2018; Liu and Blackwood, 2012; Mateo-Ortiz et al., 2014a; Ortega-Zúñiga et al., 2019; Pauli et al., 2019; Šašić et al., 2015; Sierra-Vega et al., 2019, 2018; Wahl et al., 2014). There are numerous advances beyond those described in a previous review by Xie and Puri, which focused on the transfer of material to the dies, specifically on feed shoe studies (Xie and Puri, 2006).

The objective of this review is to give a comprehensive and critical overview focusing mainly on the second stage of die filling: the powder flow promoted by the paddle wheel in a confined space. Advances in the understanding of powder flow behavior and the application of residence time distribution (RTD) theory within the feed frame, the effect of the paddle wheel design and process parameters in feed frame on critical quality attributes (CQA) of the tablets (tablet weight and weight variability), and the use of feed frame within Process Analytical Technology (PAT) system are presented.

2. Powder flow behavior within the feed frame

The understanding of powder flow behavior is essential to guarantee the success of pharmaceutical operations in the production of tablets (Dülle et al., 2018a; Escotet-Espinoza et al., 2018c; Peeters et al., 2015; Polizzi et al., 2016; Portillo et al., 2010, 2009). The powder flow behavior is dependent on many factors, such as operating conditions and design parameters of the pharmaceutical unit, and material properties (i.e. particle size distribution, particle shape, moisture content, cohesion). The feed frame can alter the particle size distribution, change the powder flow properties, cause remixing and undesirable phenomena, such as segregation and powder over-lubrication (Dalvi et al., 2018; Dülle et al., 2018a; Hildebrandt et al., 2019a; Ketterhagen, 2015; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016; Mendez et al., 2010; Méndez et al., 2012a, 2012b; Peeters et al., 2016; Sierra-Vega et al., 2019; Van Snick et al., 2018; Wahl et al., 2014).

This section presents the computational and experimental studies carried out to improve the understanding of powder flow behavior within the feed frame. DEM has been an essential tool to elucidate the powder flow behavior within the feed frame. Discrete element method was developed in 1979 (Cundall and Strack, 1979) and is now frequently used in the pharmaceutical industry to understand particulate systems (Alizadeh et al., 2014; Escotet-Espinoza et al., 2018a; Guo et al., 2011; Hildebrandt et al., 2019b, 2018; Ramírez-Aragón et al., 2018).

2.1. Wave behavior

The movement of the feed frame paddles generates a backward flow, creating crests and troughs on the powder surface during powder displacement, which is called wave behavior (Stewart et al., 2001). The wave behavior promotes rearrangement of particles, inducing segregation by percolation within the feed frame as demonstrated by DEM simulations (Mateo-Ortiz et al., 2014b). The powder wave frequency is proportional to the paddle wheel speed (Liu and Blackwood, 2012; Mateo-Ortiz and Méndez, 2016), and the powder wave amplitude depends on the gap between the top of the paddle and the top of the feed frame (Mateo-Ortiz and Méndez, 2016). The wave behavior decreases as the number of paddles is increased, causing a lower die weight variability, as the empty space behind the moving paddle is reduced (Mateo-Ortiz and Méndez, 2016). However, these last observations have been not experimentally demonstrated.

Fig. 3(a) and (b) show the wave behavior observed in DEM simulations at 60 rpm paddle wheel speed and two paddle heights; 6.5 and 4.5 mm, respectively. These figures show that as the paddle height increases, the wave behavior is limited (Mateo-Ortiz and Méndez, 2016). Fig. 4 shows the wave behavior observed experimentally within a two-chamber feed frame from a Manesty Betapress tablet press at 24 rpm paddle wheel speed. The crest and the trough can also be clearly identified.

Recent studies have considered wave behavior when performing in-line monitoring in the feed frame, because of its effect on powder blend uniformity, the estimated effective sample size within the feed frame, as well as the measurements and performance of the near-infrared (NIR) calibration model. In-line monitoring within the feed frame has been performed using NIR spectroscopy, NIR chemical imaging, Raman spectroscopy, light-induced fluorescence (LIFS) and red, green and blue (RGB) color imaging. The wave behavior may cause variation in the NIR spectral baseline (De Leersnyder et al., 2018; Mateo-Ortiz et al., 2014a; Sierra-Vega et al., 2018). The spectral baseline can provide useful information regarding the process dynamic and the physical properties of the materials (Martínez et al., 2013; Ortega-Zúñiga et al., 2019; Ropero et al., 2011, 2009; Sierra-Vega et al., 2019, 2018). In fact, experimental studies have demonstrated that at high paddle wheel speed the baseline variation of the NIR spectra acquired in-line within the feed frame is smaller (Mateo-Ortiz et al., 2014a; Sierra-Vega et al., 2018). This observation is due to a powder flat top surface within the feed frame, which means that at high paddle wheel speed, the wave behavior tends to disappear. However, the die disc speed does not significantly affect wave behavior (Sierra-Vega et al., 2018). A recent study has proven that cutting notches of 1 mm deep, into the paddles of the wheel in the position where the NIR probe acquires the spectra, is sufficient to significantly reduce the variations in absorbance at one wavelength (1300 nm) of NIR spectra (De Leersnyder et al., 2018). This study also investigated the effect of distance between the probe and the paddle top on the absorbance at 1300 nm and found that 3 mm was enough to observe a high variation caused by wave behavior.

The powder wave behavior within the feed frame can affect the tablet weight variability; therefore, it is an aspect to be considered during tablet manufacturing. Increasing the number of paddles (De Leersnyder et al., 2018; Mateo-Ortiz and Méndez, 2016) and the paddle wheel speed (Mateo-Ortiz et al., 2014a; Sierra-Vega et al., 2018) can reduce the powder wave behavior. Nevertheless, at high paddle wheel speed the powder blends, which are prone to attrition or over-lubrication can be affected.

2.2. Velocity and force gradients

DEM simulations have been used successfully in the study of particle velocity profile within the feed frame (Hildebrandt et al., 2019a; Ketterhagen, 2015; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016). The particle velocity profiles are generated by the different tangential velocities of particles. The velocity gradient is higher when the paddle wheel speed is higher (Ketterhagen, 2015; Mateo-Ortiz et al., 2014b). The velocity of the particles farthest from the rotation axis is greater than particles closer to it (Ketterhagen, 2015; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016). Therefore, increasing the diameter of the paddle wheel center (hub) reduces the velocity gradient between the particles (Ketterhagen, 2015). Specifically, in a DEM simulation of a one-chamber feed frame, the particles near the hub reached speeds of less than 50% of the paddle wheel speed (Mateo-Ortiz and Méndez, 2016). Similarly, DEM was used to study the velocity profiles within a three-chamber feed frame (Hildebrandt et al., 2019a), founding that the mean particle speed in the distributing chamber and filling wheel chamber had mean values slightly lower than the paddle wheel speed. However, *surprisingly,* this study also reported that the overall mean speed of the particles in the dosing wheel chamber was 1.9 times higher than paddle wheel speed. The authors attributed this behavior to three mechanisms: first, a high momentum transfer to the particles from the rectangular paddles, second, the number of large

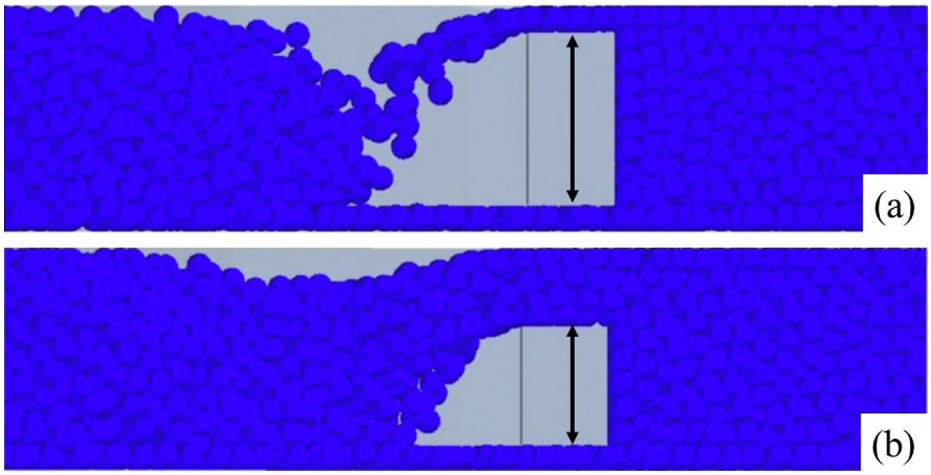


Fig. 3. Powder flow behavior within the feed frame (a) 6.5 mm paddle height (b) 4.5 mm paddle height (Reprinted from reference (Mateo-Ortiz and Méndez, 2016)).

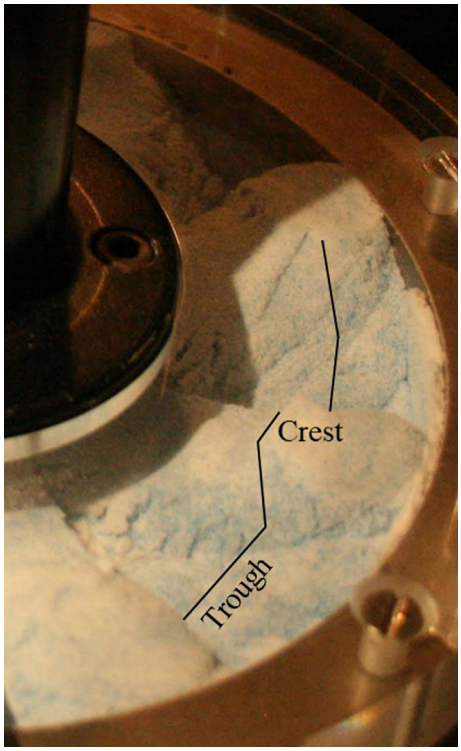


Fig. 4. Powder flow behavior observed experimentally within the feed frame from Manesty Betapress tablet press at paddle wheel speed of 24 rpm (Adapted from reference (R. Mendez et al., 2010)).

particles found in the dosing chamber and third, the high void volume between the paddles. More studies are necessary to evaluate the particle velocity profiles within this chamber.

The force exerted (sum of tangential and normal forces) by the paddle on the particles has also been studied using DEM (Ketterhagen, 2015; Mateo-Ortiz and Méndez, 2015). The force exerted on the particles is higher when either the paddle wheel speed or the particle radial distance increases (Ketterhagen, 2015; Mateo-Ortiz and Méndez, 2015). However, the force exerted decreases as the particles move away from the paddle, where the gravity force starts having a more significant impact (Ketterhagen, 2015). Also, the effect of particle cohesion on work done was investigated using DEM, showing that the work done by the paddle increases with the paddle wheel speed and the cohesion level (Ketterhagen, 2015). The work done on cohesive particles increases to overcome the interaction force between the particles (Ketterhagen, 2015).

The velocity and force profiles can cause changes in the consolidation of the material within the feed frame, negatively affecting the die filling process and therefore, the CQA of the tablets. Also, the higher the force exerted by the paddle on the particles, the higher the probability of particle attrition. However, these profiles can be reduced by increasing the number of paddles and the diameter of the center of the paddle wheel.

2.3. Mass steady-state and mass hold-up

The mass steady-state within the feed frame indicates that the powder mass flow that enters is equal to the mass of the tablets that are manufactured per unit of time. Computational studies demonstrated that an increase in the paddle wheel speed decreases the time to reach mass steady-state (Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016). These results were confirmed experimentally using the baseline changes of NIR spectra acquired in-line in the feed frame (Sierra-Vega et al., 2018). This study also demonstrated that NIR spectral baseline variability was more considerable during start-up and at the end of the process, but minimal during the mass steady-state. The mass steady-state was also studied through DEM simulations (but not experimentally) as a function of the paddle height and the number of paddles, demonstrating that these factors do not have a significant effect on the time to reach mass steady-state (Mateo-Ortiz and Méndez, 2016).

Studies show that DEM can also be utilized to evaluate the mass hold-up within the feed frame (Hildebrandt et al., 2019a, 2018; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016). The mass hold-up is the quantity of powder available within the feed frame for die filling, once the feed frame reaches the mass steady-state. The mass hold-up within the feed frame depends directly on the die disc speed, which regulates the outlet flowrate (Hildebrandt et al., 2019a; Mateo-Ortiz et al., 2014b). At the same die disc speed, the mass hold-up remains approximately constant (Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016).

Variations in the mass hold-up or feed frame filling level affect the baseline of the NIR spectra acquired in-line in the feed frame (Mateo-Ortiz et al., 2014a; Sierra-Vega et al., 2018), due to a decrease in the fraction of radiation that returns to the detector when the distance between the probe and powder bed increases (Ropero et al., 2009; Sánchez-Paternina et al., 2016). These variations in the NIR spectral baseline have also been used to determine the fill level in the feed frame and help monitor process dynamics (Mateo-Ortiz et al., 2014a; Sierra-Vega et al., 2018).

An experimental study showed that for a material with poor flow properties (compressibility index (%): 18.09 ± 0.31), the paddle wheel

speed (20 – 140 rpm), die disc speed (250 – 1000 tablet per minutes) and fill depth (8–12 mm) do not have a significant effect on the mass hold-up (Peeters et al., 2015). However, the mass hold-up was significantly affected by paddle wheel speed and die disc speed for material with good flow properties (compressibility index (%): 7.41 ± 0.56).

The mass hold-up was also recently determined using an optical triangulation sensor (an ultra-high-speed and high accuracy laser displacement sensor) to monitor the powder level within the filling chamber of a three-chamber feed frame (Dühlmeyer et al., 2019a, 2019b, 2018). The filling level was more significant near the end of the paddle, and lower when approaching the paddle wheel center when a material with excellent flow properties was used; likewise, the filling level increases at high paddle wheel speed (Dühlmeyer et al., 2018). This because the centrifugal forces caused by the rotational movement, which led to a radial displacement of the powder to the edge region of the chamber (Dühlmeyer et al., 2018). The results of this study can be linked to the velocity gradients within the feed frame, described in Section 2.2. The optical triangulation sensor was also used to compare the influence of two different paddle wheel designs (round and flat paddle wheels) on the level of a poor flowing material within the filling chamber (Dühlmeyer et al., 2019a). The round paddle increased the powder level and the mass powder within the filling chamber. This behavior was linked to number of paddles, the gap between the paddle bottom and feed frame bottom, and to the distribution of the paddles in the round paddle wheel. A subsequent study demonstrated that an increase in the die disc speed reduced the powder level within the filling chamber regardless of material properties; nevertheless, this study also reported that the paddle wheel speed had a positive effect on filling level for poor flowing powder, whereas it had a negative effect for powder with good flow properties (Dühlmeyer et al., 2019b). In these studies, the volume of the feed frame was reduced by placing an acrylic lid instead of a dosing wheel, limiting the application of these studies for real tablet manufacturing processes.

These studies clearly show the opportunity for real-time monitoring of process dynamics within the feed frame. The close monitoring of the NIR spectral baseline or filling level is an opportunity for the detection of disturbances that could affect tablet quality. Also, the monitoring of the filling level provides information that can be used to optimize the paddle wheel design to improve the powder distribution within the feed frame.

2.4. Particle attrition and powder lubrication

A reduction in particle attrition in pharmaceutical powder blends leads to an improvement in tablet quality (Mehrotra et al., 2007). The powder attrition within the feed frame is related to the force exerted to the particles and depends on the initial particle size distribution (PSD) and the particle strength (Ketterhagen, 2015). To study the effect of different feed frames on the PSD, a Manesty Beta Press feed frame and a Fette 3090 feed frame using a simulated die disc were used (Méndez et al., 2012b). The powder presented a reduction in the particle size after passing through the feed frames; however, the three-chamber feed frame showed a more significant effect. Also, this study showed that the granular materials were more prone to breakage within the feed frame. These studies were extended to high-speed rotary tablet presses where a significant decrease in particle size was also reported (Peeters et al., 2015).

DEM was also used to investigate the potential for attrition. As the particle size increases, greater tangential and compressive forces are applied, which potentially increase the risk of breakage (Mateo-Ortiz and Méndez, 2015). The powder attrition can be reduced by increasing the hub diameter of the paddle wheel, and using a wheel with angled paddles (Ketterhagen, 2015). The particle attrition could lead to changes in the bulk and tap density of the powder within the feed frame (Méndez et al., 2012b, 2012a), these changes can produce variability in the properties of the tablets, such as tablet weight, hardness and dissolution (Román-Ospino et al., 2016).

Recent studies have also addressed the effect of powder lubrication on powder behavior within the feed frame (Dülle et al., 2018a; Ketterhagen, 2015; Mateo-Ortiz and Méndez, 2015; Méndez et al., 2012a, 2012b, 2010; Peeters et al., 2016; Perrault et al., 2011). Pharmaceutical formulations are lubricated to reduce die-wall friction, improve the powder flow properties, increase the bulk density and prevent the sticking of material to the process equipment (Aly, 2006; Lakio et al., 2013; Mehrotra et al., 2007). However, large shear on lubricated powder blends can adversely affect the properties of the powder blends and tablets (Hernandez et al., 2016; Méndez et al., 2012a). The intensity of over-lubrication effects within the feed frame depend on the concentration of lubricant (magnesium stearate, MgSt) in the powder blends before entering the feed frame, and on the paddle wheel speed (Méndez et al., 2012a, 2010; Perrault et al., 2011). Research on this topic has shown that as the paddle wheel speed increases, the cohesion and the dilation for the powder blends containing MgSt are reduced; however, on non-lubricated powder blends, the paddle wheel speed does not induce significant changes in their flow properties (Méndez et al., 2010). The reason for this behavior is the sensitivity to shear of powder blends that contain magnesium stearate (Mehrotra et al., 2007; Peeters et al., 2016).

Also, the influence of the paddle wheel speed on the tensile strength of the tablet containing 0.5% w/w of MgSt and prepared with powder blends with different shear levels was investigated (Peeters et al., 2016). This study highlighted that the tensile strength, for tablets prepared with high-shear blending without MgSt, was not affected by the paddle wheel speed. In contrast, the tensile strength of tablets prepared with low-shear blending decreased as paddle wheel speed increased. Powder lubrication was also studied as a function of the volume of the feed frame using a two-chamber and a three-chamber feed frame, revealing that the shear applied was proportional to the volume of the feed frame (Méndez et al., 2012b). Also, the shear increases the powder hydrophobicity, which in turn, decreases the hardness and dissolution rate of the tablets (Méndez et al., 2012a).

The shear applied on the powder blends within the feed frame has been quantified by the number of paddle passes using Eq. (1) (Dülle et al., 2018a; Ketterhagen, 2015; Mateo-Ortiz and Méndez, 2015; Méndez et al., 2010). This equation expresses that the number of paddle passes is a function of paddle wheel speed, the number of paddles, and the mean residence time (MRT) of the powder within the feed frame. MRT represents the average time the material spent within the feed frame.

$$Paddle\ \ passes = Paddle\ \ wheel\ \ speed(RPM) * number\ \ of\ \ paddles * MRT(min) \quad (1)$$

Eq. (1) shows a proportional relationship between the number of paddle passes and the paddle wheel speed. However, experimental studies have demonstrated that at a constant number of paddles, an increase in paddle wheel speed with a simultaneous increase in die disc speed does not affect the number of paddle passes (Dülle et al., 2018a), because of a significant reduction in MRT when the die disc speed increases. Similar observations were obtained in computational studies (Ketterhagen, 2015; Mateo-Ortiz and Méndez, 2015).

2.5. Segregation

Segregation has been reported in different pharmaceutical process units, such as hoppers (Anand et al., 2009; Ketterhagen et al., 2008), rotating cylinders (Khakhar et al., 2003), fluidized beds (Wormsbecker et al., 2005), and transitional chutes (Bhattacharya and McCarthy, 2014; Jaklič et al., 2015). The segregation of the pharmaceutical powder is based on four mechanisms: percolation, angle of repose segregation, trajectory segregation, and fluidization segregation

(Ketterhagen et al., 2008). DEM simulations and experimental studies have demonstrated that percolation is the most significant segregation mechanism within the feed frame, localizing the large particles mainly in the top of the feed frame and the small particles in the bottom (Dülle et al., 2018a; Hildebrandt et al., 2019a; Mateo-Ortiz et al., 2014b). Also, these studies have shown that particle size segregation is transferred into the dies, affecting the active pharmaceutical ingredient (API) distribution in the tablets, placing the larger particles mostly at the top surface of the tablet (Dülle et al., 2018a).

DEM simulation was also used to evaluate segregation within each of the three chambers of a Fette 1200i feed frame employing particles between 1 and 2.8 mm in diameter (Hildebrandt et al., 2019a). The three chambers in the feed frame promoted particle size segregation. Nonetheless, the two bottom chambers contributed more to the global segregation within the feed frame, because the materials fall from distributing chamber with a degree of segregation. The importance of the paddle wheel speed on segregation was also demonstrated using DEM. At low paddle wheel speed, the particle size segregation is promoted in formulations susceptible to segregation (Hildebrandt et al., 2019a; Mateo-Ortiz et al., 2014b), because the particles have more time for rearrangement and the smaller particles reach the bottom easily (Dülle et al., 2018a; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2015).

Segregation can affect the performance of in-line monitoring within the feed frame. Generally, the spectroscopic probe is located on the feed frame and analyzes approximately to a depth of 1-mm. If particle size segregation occurs, there is a sub-estimation or over-estimation of the drug concentration, depending on whether the API or some excipient is segregated. This is because the in-line analysis is limited to the depth of penetration of the analytical technique (Bakri et al., 2015; Berntsson et al., 1999; Huang et al., 2016; Ortega-Zúñiga et al., 2019; Román-Ospino et al., 2018). NIR spectroscopy was used to monitor drug concentration within the feed frame while segregation occurred (Wahl et al., 2014). This study found an accumulation of API toward the process end, producing tablets with high drug content, suggesting that the factors that possibly influenced the segregation were: hopper cross-sectional shape, hopper angle, and vibration of the tablet press. Similarly, near-infrared chemical imaging (NIR CI) was used to monitor the feed frame (Dalvi et al., 2018). The chemical images were divided into four zones, and different predicted API concentrations were obtained in each zone. These differences between the local concentrations were linked to segregation within the feed frame.

Fig. 5(a) illustrates the segregation within the feed frame, the small particles (blue particles) in the bottom and large particles (red particles) mainly in the top (Mateo-Ortiz et al., 2014b). Fig. 5(b) shows the segregation in tablets prepared with Vivapur®102 (d_{50} = 131 µm) and a blue tracer (400–315 µm) (Dülle et al., 2018a). Clearly, large particles (tracer) are observed on the upper tablet surface and the smaller particles in the lower tablet surface. The smallest particles reach the dies first, due to the segregation in the feed frame (Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2015).

Particle size segregation within the feed frame can be reduced using materials with similar particle size distribution. However, the paddle wheel speed has a vital role in segregation within the feed frame (Ketterhagen, 2015; Mateo-Ortiz et al., 2014b; Mateo-Ortiz and Méndez, 2016). An increase in paddle wheel speed increases the degree of consolidation of powder which avoids particle rearrangement (Mateo-Ortiz and Méndez, 2016). Also, to study the segregation within the feed frame, it is necessary to determine the distribution of the materials at the entrance. The concentric tablet press hopper can induce more segregation in the powder that is entering the feed frame than an eccentric hopper (Ketterhagen and Hancock, 2010).

2.6. Mixing effects

Although the feed frame can cause undesirable phenomena on the properties of the powder blends, it has also been reported as an additional mixing element. Both macroscopic and microscopic mixing can occur within the feed frame (Van Snick et al., 2018). The macroscopic mixing effect within the feed frame helps to reduce the powder blend heterogeneity (Van Snick et al., 2018). The microscopic mixing within the feed frame could cause over-lubrication and affect the tablet properties (Méndez et al., 2012a; Peeters et al., 2015). The studies of the mixing effect within the feed frame have been carried out using different methodologies: visualizing of the powder flow pattern, measuring the in-line drug concentration before and within the tablet press feed frame, and determining to mean centered variance (MCV) within a study of residence time. In this section, the studies made with the first two methodologies will be presented, since the determination of the residence time is the topic of **Section 3**.

The flow patterns within the feed frame from the Manesty Betapress tablet press were determined by addition of blue powder to the material entering the feed frame and acquiring images with a digital camera (Méndez et al., 2010). The blue powder formed a circular pattern in the first chamber before moving to the second chamber, but after approximately one minute the second chamber presented a more uniform color, indicating a significant mixing effect compared to the first chamber. DEM simulations have been used to investigate the mixing effect by the flow patterns. The powder flow patterns are extremely dependent on the paddle wheel speed. High paddle wheel speeds generate a better distribution of incoming materials within the feed frame (Ketterhagen, 2015).

A first-order kinetic model was used to determine the transition dynamics between step changes performed in the two-chamber feed frame (Dalvi et al., 2019; Gosselin et al., 2017). This model showed that the first-order rate constants varied between the different transitions, which was linked to the mixing of the incoming material and the material present in the feed frame, as well as to the flow properties of the powder blends. The mixing effect within the feed frame also depends on the material properties.

The mixing effect within the feed frame was demonstrated by a high correlation between the drug concentration in the feed frame and the tablet potency by using a series of step changes and monitoring these changes with in-line NIR spectroscopy (Ward et al., 2013). Similarly, the drug concentration of powder blend in the chute located after the blender, and in the tablet press feed frame of a continuous direct compaction tablet manufacturing line were compared with the drug concentration of tablets (Sierra-Vega et al., 2019). The relative standard deviation of the drug concentration was reduced by approximately 60% in the feed frame, when compared to the chute. The drug concentration within the feed frame and the tablets did not show significant differences, contrary to the results when the drug concentration of tablets was compared to drug concentration in the chute. These results were attributed to the mixing effect and to reduction in sampling and analytical errors within the feed frame.

The additional mixing is a desired effect within the feed frame to reduce the probability of rejecting a powder blend. The additional mixing will depend on the heterogeneity with which the powder blend enters the feed frame and on the paddle wheel speed (Ketterhagen, 2015; Sierra-Vega et al., 2019). The higher the heterogeneity of the blend, the greater the mixing effect within the feed frame on it. At high paddle wheel speed, a more significant mixing effect is also expected (Dülle et al., 2018b; Ketterhagen, 2015); however, these conditions also increase the risk of particle attrition and over-lubrication for formulations susceptible to these effects.

3. RTD in the feed frame

The residence time distribution is the probability distribution of time that solid or fluid materials spend within one or more process units in a continuous system (Gao et al., 2012). The RTD is used for traceability of materials in a continuous system, to design equipment,

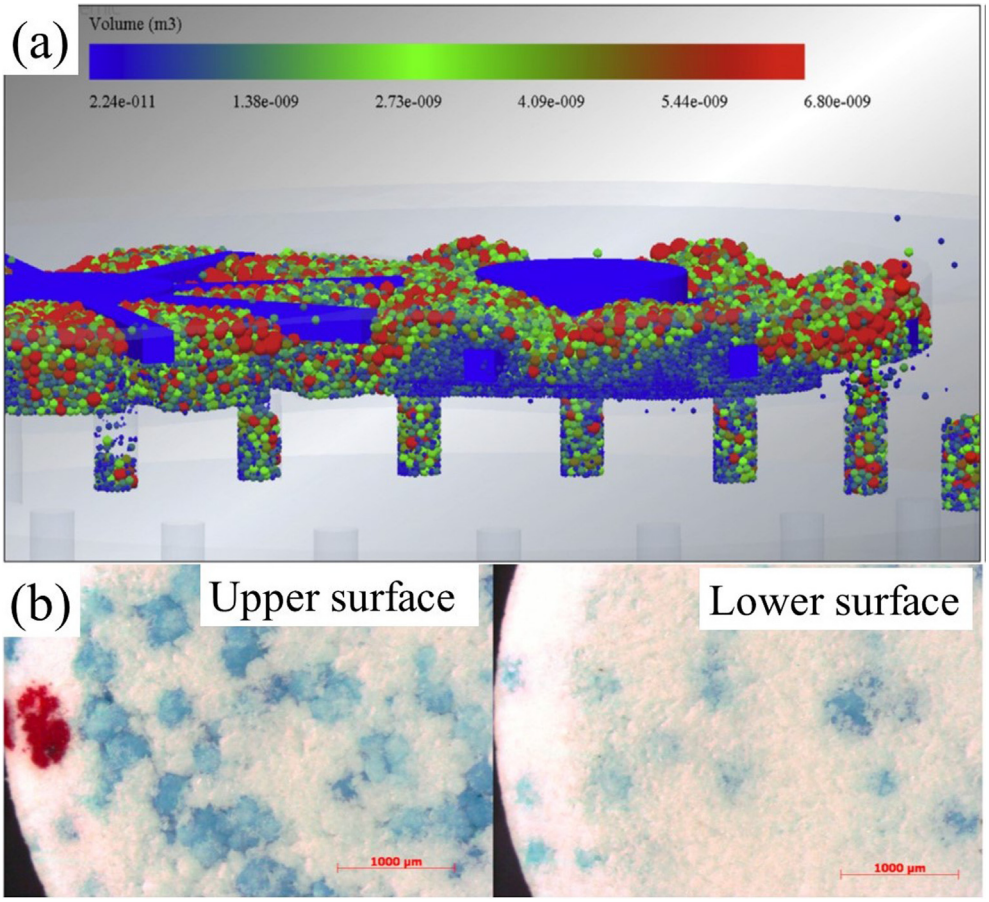


Fig. 5. Segregation (a) within the feed frame (Reprinted from reference (Mateo-Ortiz et al., 2014b)) (b) in tablets (Reprinted from reference (Dülle et al., 2018a)).

measure and upgrade the performance of the units, and troubleshooting (Engisch and Muzzio, 2016; Nauman, 2008). The RTD in unit operations for solids depends on the design parameters, operating conditions and material properties (Escotet-Espinoza et al., 2019, 2018b; Gao et al., 2011; Portillo et al., 2010; Vanarase et al., 2013).

Fig. 6 presents a flowchart that summarizes the procedure followed to obtain the RTD measurements in solid processing units using step input or pulse input (Dülle et al., 2019a, 2019b, 2018a, 2018b; Engisch and Muzzio, 2016; Fogler, 2016; Mateo-Ortiz and Méndez, 2015; Osorio et al., 2016; Portillo et al., 2009; Vanarase and Muzzio, 2011). To begin the RTD measurement, the continuous system is turned on and operated until it reaches the mass steady-state. In the feed frame, the time to reach mass steady-state depends on the paddle wheel speed (Mateo-Ortiz et al., 2014b; Sierra-Vega et al., 2018). After having achieved mass steady-state, a tracer is injected to the input stream of the process unit. The most common methods to inject the tracer is a pulse input or a step input (Fogler, 2016; Gao et al., 2012). In powder processing unit operations, the pulse input is used to characterize the process unit dynamics, and the step input is used for traceability of the materials. In a pulse injection, the tracer concentration is measured at the output of the system for a time that ensures that the tracer has left completely. In a step input, the tracer concentration is measured at the output until the effluent reaches stability in concentration. A concentration vs. time profile is obtained at the output of the system and different fit models can be used to adjust this profile. The stirred tank in series model, Taylor dispersion model, and convolution model have been used to adjust the concentration profile in residence time measurement in unit operations for solids (Engisch and Muzzio, 2016; Gao et al., 2012; Mateo-Ortiz and Méndez, 2015). The use of these fit models minimizes the noise from the experimental RTD measurements. The concentration profile allows calculation of the RTD function, *E(t)*, which represents the fraction of material leaving the system at each time. Based on *E(t)*, the MRT and MCV are calculated. MRT gives the average time the material spent in the process unit and the MCV indicates the degree of dispersion of the tracer within the process unit. The *E(t)*, MRT and MCV can be calculated using the following mathematical relationships (Fogler, 2016). The calculation of *E(t)* depends on the type of injection performed.

$$E(t)_{step} = \frac{d}{dt}\left[\frac{C(t)}{C_0}\right] \tag{2}$$

$$E(t)_{Pulse} = \frac{C(t)}{\int_0^\infty C(t)dt} \tag{3}$$

$$MRT = \tau = \int_0^\infty tE(t)dt \tag{4}$$

$$MCV = \int_0^\infty (t-\tau)^2 E(t)dt \tag{5}$$

Computational and experimental studies coincide that an increase in die disc speed decreases the MRT due to an increase in the outlet mass flow rate (De Leersnyder et al., 2019; Dülle et al., 2019a, 2018b, 2018a; Mateo-Ortiz and Méndez, 2015; Méndez et al., 2010). The RTD was also investigated as a function of paddle wheel design, particle cohesion, and particle size using DEM. At paddle wheel speed of 10 rpm, the paddle wheel design and particle cohesion do not have a significant effect on the RTD (Ketterhagen, 2015).

The effect of particle size on RTD has not been fully understood. A computational study showed that the larger particles have a longer residence time within the feed frame, as a result of a particle size segregation (Mateo-Ortiz et al., 2014b). Nevertheless, an additional study demonstrated that the MRT of the particles within the feed frame does

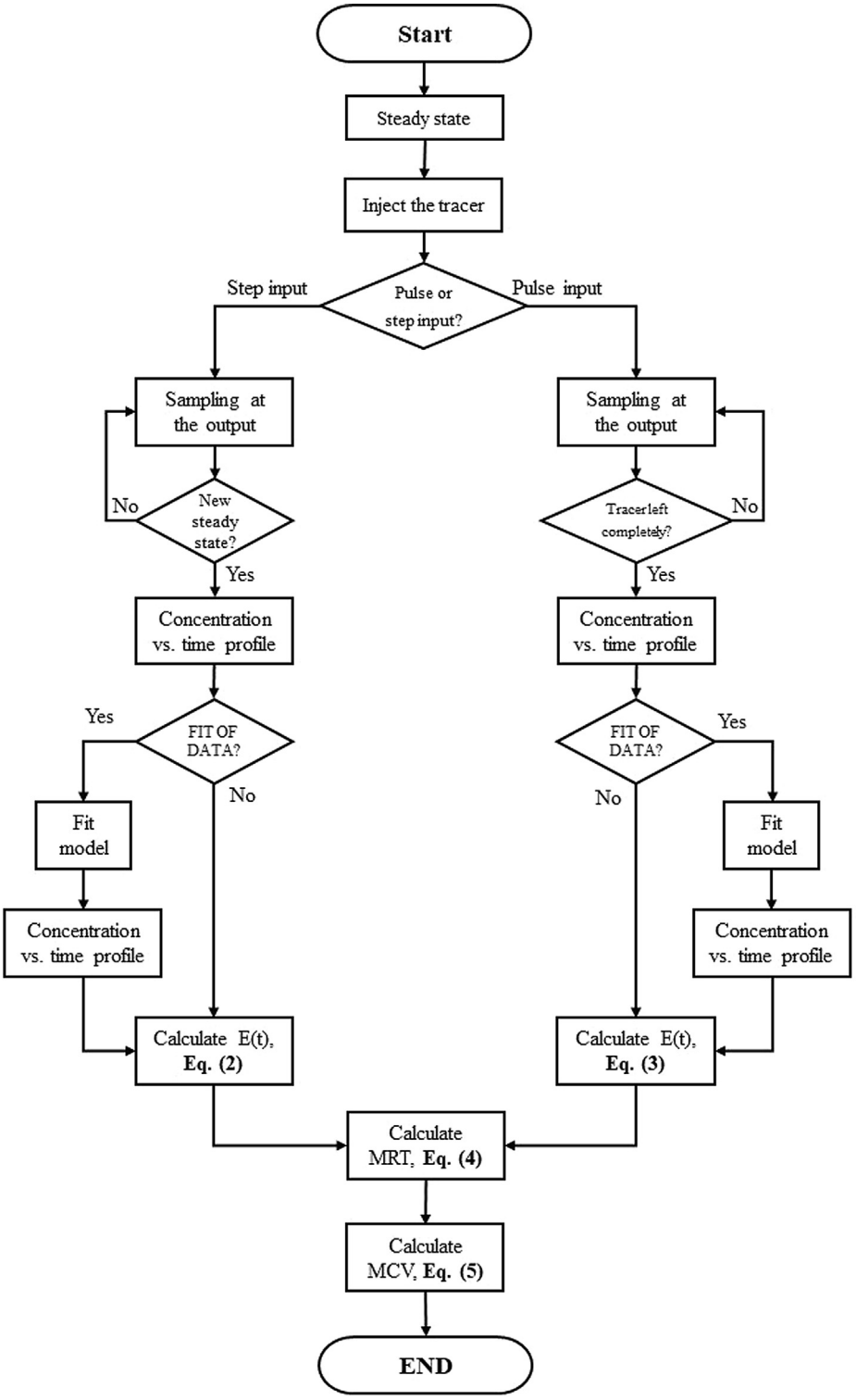


Fig. 6. Flowchart for the determination of the RTD, MRT and MCV using pulse input and step input.

not depend on their particle size (Hildebrandt et al., 2019a). This discrepancy in the results can be linked to the influence of the rather large particles on the powder flowing into the dies (Beverloo et al., 1961). Mateo-Ortiz et al. used a die diameter of 10 mm with particles up to 2.45 mm in diameter, while Hildebrandt et al. used a die diameter of 24 mm with particles up to 2.8 mm in diameter. A smaller flow area limits the entry of large particles to the dies, which increases the residence time. Additional studies are necessary to understand the effect of particle size segregation on the measurement of residence time distribution.

The RTD has been determined experimentally using a feed frame with a simulated die disc, and a feed frame mounted on a tablet press. These studies showed that an increase in the die size and in the filling depth causes a decrease in the MRT (De Leersnyder et al., 2019; Dülle et al., 2019b, 2018b, 2018a; Méndez et al., 2010). However, an increase in the acetaminophen concentration in powder blends causes an increase in MRT (Méndez et al., 2010). This behavior is because the powder blend cohesion increases, causing a reduction in the powder

mass in the die. The effect of two different geometries of feed frame (two-chamber and three-chamber) on MRT was also studied (Méndez et al., 2012b). The MRT within the feed frame increases with the volume of the feed frame. Similar results were found when reducing the volume of a three-chamber feed frame with a Plexiglas disc in one of the bottom chambers (Dülle et al., 2019b).

Residence time distribution was also determined in a Fette 102i tablet press (Dülle et al., 2019b, 2019a, 2018b, 2018a). 160 g of the tracer were added as a layer into the feed pipe, and its concentration was determined in the tablets. These studies demonstrated that the MRT is not affected by the paddle wheel speeds. In contrast, a DEM simulation showed that an increase in the paddle wheel speed decreases the MRT (Mateo-Ortiz and Méndez, 2015). In addition to the type of the studies (computational and experimental), the differences between simulated and experimental results can be linked to the paddle wheel speeds used. The simulations reported that an increase in paddle wheel speed (from 24 to 72 rpm) increases the powder mass in the dies, thereby decreasing the MRT, whereas in experimental studies, the increase in the paddle wheel speed (from 40 to 120 rpm) did not report variation in the tablet weight, meaning that at 40 rpm, the dies were filled entirely and a change in the paddle wheel speed did not affect its weight. Also, these studies demonstrated that the MCV increased as the paddle wheel speed increased, consistent with computational results (Ketterhagen, 2015). This observation is essential to understand the mixing effects (Section 2.6) within the feed frame at high paddle wheel speed. At high paddle wheel speed, the greater the degree of dispersion of the powder within the feed frame.

The effect of the design parameters of the filling paddle wheel on RTD was also studied. The MRT and MCV increased slightly when a wheel with round paddles was used instead of one with flat paddles (Dülle et al., 2019b). The MRT was not affected by an increase in the hub diameter of the paddle wheel; however, the MCV decreased with a large hub (Dülle et al., 2019a). The gap size between the feed frame and the die disc did not significantly affect the RTD (Dülle et al., 2019a). Also, the MRT decreased when an outer scraper (avoid the effect of recirculation of the scraped powder) was used (Dülle et al., 2018b). Additionally, when the amount of tracer added increased, the MRT and MCV increased. However, the residual moisture content did not have a significant effect on these RTD parameters (Dülle et al., 2018a).

In-line NIR spectroscopy measurement was also used to determine the residence time distribution, and its results were compared with those obtained from the produced tablets (De Leersnyder et al., 2019). The NIR probe was placed in the top and bottom of the feed frame; the RTD profiles obtained from bottom measurements showed a more considerable agreement with the RTD profiles obtained from the tablets. This behavior was linked to the existence of layers with different concentrations within the feed frame.

The application of the residence time theory allows a better understanding of the powder flow behavior and the mixing effect within the feed frame, helps the scale-up process, and improves the feed frame design (Dülle et al., 2018b; Gao et al., 2012; Mateo-Ortiz and Méndez, 2015). For a formulation, the RTD within the feed frame depends mainly on the paddle wheel speed and the die disc speed. A low MRT is desirable to reduce the impact of feed frame on the CQA of the tablets. However, the MRT within the feed frame must guarantee an adequate die filling and minimize adverse effects such as segregation, over-lubrication, or powder attrition, which can affect both powder blends and tablet quality. This section evidences the need to study the effect of particle size segregation on the measurement of residence time distribution.

4. Effect of the paddle wheel design and feed frame process parameters on tablet weight

The tablet weight is a critical quality attribute that must be controlled to guarantee the amount of drug that a patient receives (FDA, 2004; Goodwin et al., 2018). The tablet weight is mainly affected by die disc speed (turret speed or tableting speed), the paddle wheel speed, and the feed frame paddle wheel design (Dühlmeyer et al., 2019a; Goh et al., 2018b; Grymonpré et al., 2018b, 2018a; Peeters et al., 2015; Van Snick et al., 2018; Yaginuma et al., 2007). This section presents the studies that have investigated the influence of the paddle wheel speed, die disc speed and paddle wheel design on the weight and weight variability of the tablets. The studies considered in this section are those directly developed on the rotary tablet press, where all possible variables which might affect the die filling are involved.

Different studies agree that an increase in paddle wheel speed leads to an increase in tablet weight, especially when powder blends with poor flow properties are used (Grymonpré et al., 2018b; Peeters et al., 2015; Van Snick et al., 2018), because an increases the paddle wheel speeds, increases the normal force exerted by the paddle on the material, forcing it to enter the dies (Mateo-Ortiz and Méndez, 2015). However, this effect is not significant at high paddle wheel speeds, where the powder mass in the dies remains constant due to that the force applied on the powder in the filling zone do not increased the powder consolidation within the dies. This last behavior was confirmed by different studies, in which an increase in paddle wheel speed from 40 to 120 rpm did not affect the weight of microcrystalline cellulose (Vivapur®102) tablets (Dülle et al., 2019b, 2019a, 2018b, 2018a). Moreover, an increase in die disc speed causes a decrease in tablet weight and an increase in tablet weight variability (Grymonpré et al., 2018a; Peeters et al., 2015; Van Snick et al., 2018; Yaginuma et al., 2007). It is reasonable to consider that the tablet weight is affected by the die disc speed, because an increase in the die disc speed reduces the time that the die stays within the die fill region, causing a non-uniform powder distribution within the dies (Mehrotra et al., 2009; Van Snick et al., 2018; Yaginuma et al., 2007).

Tablet weight and tablet weight variability have also been studied as a function of the die fill depth (8–12 mm), gap size between the feed frame and the die disc (11 and 25 mm), the overfill level and the deaeration in the feed frame (Dülle et al., 2019b; Grymonpré et al., 2018b; Peeters et al., 2015). It was found that a higher die fill depth increased the die volume and therefore, the tablet weight increased, but the tablet weight variability was not affected by the increase in die fill depth (Peeters et al., 2015). Likewise, as the gap size increased, the accumulation of material on the dies also increased, thereby increasing the tablet weight and reducing the tablet weight variability (Dülle et al., 2019a). Finally, the overfill level showed a positive effect on tablet weight. On the other hand, the deaeration in the feed frame did not show a significant effect (Grymonpré et al., 2018b).

The paddle wheel design also plays an essential role in defining the tablet weight and its variability when an acceptable or poorly flowing material is used (Dühlmeyer et al., 2019a; Grymonpré et al., 2018b). A study used wheels with curved, flat, and round paddles to investigate the effect of paddle design on tablet weight variability (Grymonpré et al., 2018b). The results revealed that the paddle design effect on weight variability depended on paddle wheel speed and die disc speed. When paddle designs were compared as a function of paddle wheel speed, the curved paddles, and the flat paddles showed lower weight variability than round paddles. However, when this comparison was performed according to die disc speed, only the curved paddles guaranteed a low weight variability. Therefore, these results suggest that the paddle wheel speed and die disc speed should be considered in the selection of the paddle design when poor flowing materials are used. Similarly, a study investigated the influence of using a round paddle or a flat paddle in the filling chamber of a three-chamber feed frame on tablet weight and tablet weight variability (Dühlmeyer et al., 2019a). The results demonstrated that the tablets obtained with the flat paddle wheel showed a significantly lower weight and higher weight variability compared to tablets obtained with the round paddle wheel under the same conditions. Nevertheless, in this study, the volume of the feed frame was reduced by sealing the dosing chamber; therefore, the results

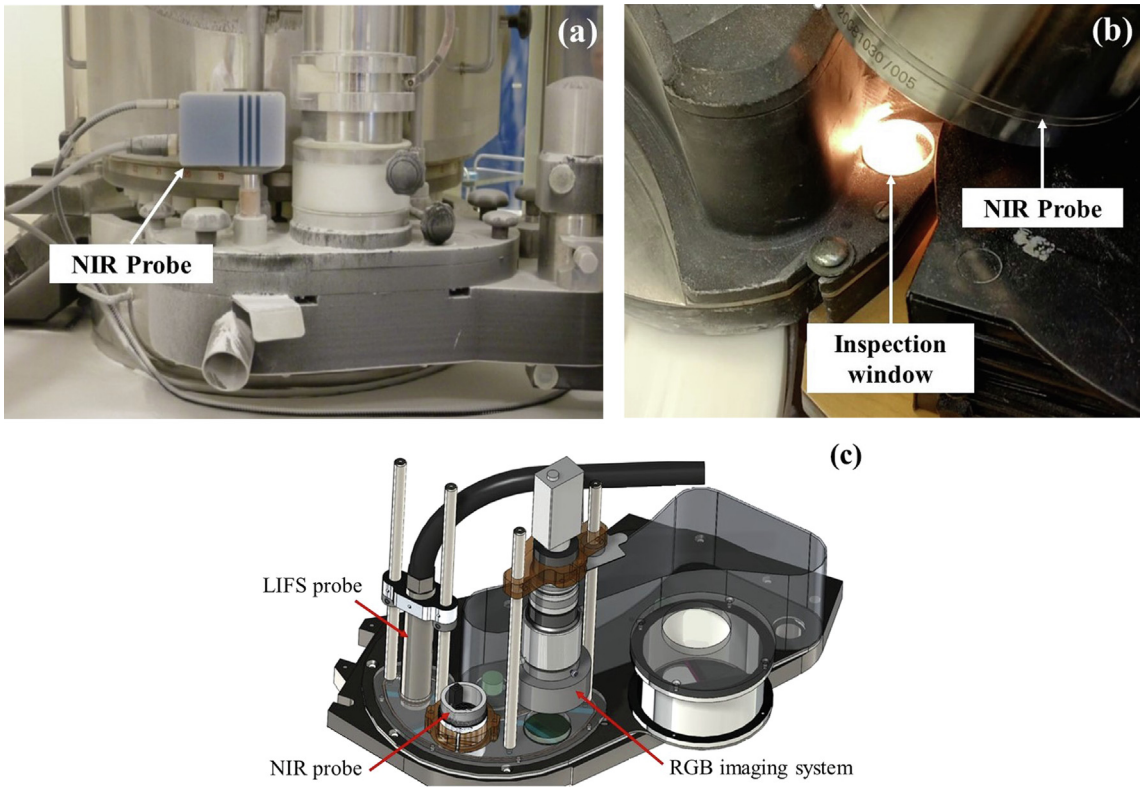


Fig. 7. Different NIR probes set up for in-line model in the feed frame (a) NIR probe installed on the tablet press (Reprinted from reference (Wahl et al., 2014)) (b) NIR probe installed on a feed frame with a simulated die disc (Adapted from reference (Ortega-Zúñiga et al., 2019)) (c) Three different PAT tools (Reprinted from reference (Durão et al., 2017)).

are difficult to extrapolate when the tablet press feed frame is operating under normal conditions, because the dosing chamber plays an essential role in the die filling process.

The effect of the process parameters on the weight and weight variability of mini-tablets of 1.8- and 3-mm diameter has also been investigated (Goh et al., 2018b). The results show that an increase in paddle wheel speed caused an increase in the mini-tablet weight, while the die disc speed had no significant effect on weight and weight variability. This die disc speed behavior on mini-tablet weight, contrary to the tablet weight, since in mini-tablets, the filling volume is relatively small and the time that the die spends within the die filling region is sufficient to fill the die.

The studies presented in this section showed that the operating conditions and paddle design must be defined based on the powder blend properties to minimize the effect of the feed frame on tablet weight and tablet weight variability. For poor flowing materials, an increase in the paddle wheel speed increases the tablet weight and reduces its variability (Grymonpré et al., 2018b; Peeters et al., 2015; Yaginuma et al., 2007). Also, for these materials, it is favorable to use curved paddles to reduce the tablet weight variability (Grymonpré et al., 2018b). However, the material properties have less influence on tablet weight and its variability at low paddle wheel speeds (Van Snick et al., 2018). The selection of operating conditions and appropriate paddle design is a challenge for all mechanisms simultaneously acting within the feed frame. Therefore, the best practical method to define the operating conditions and the adequate paddle design for a specific formulation is to carry out a design of the experiment (DoE). The DoE must include the paddle wheel speed, die disc speed and paddle design as experimental factors, and tablet weight and tablet weight variability as response variables. The levels for the factors must be selected by considering prior knowledge of the compaction process or recommendations of the tablet press manufacturers. The optimization of the response variables will allow identifying the combination of the levels to reach the target weight and minimize the tablet weight variability (Montgomery, 2013). However, although these empirical studies provide excellent results, they cannot be extrapolated when using different tablet presses or different pharmaceutical formulations (Peeters et al., 2015).

5. Feed frame as PAT unit

Process Analytical Technology (PAT) has played an essential role in the modernization of the pharmaceutical industry (Lee et al., 2015; Rantanen and Khinast, 2015; Teżyk et al., 2015). PAT is focused on real-time monitoring and control of critical in-process parameters to achieve the specifications of the final product (FDA, 2004). NIR and Raman spectroscopy are non-invasive methods that have been used to monitor critical parameters within PAT systems because they are fast, accurate and suitable for real-time measurements (De Beer et al., 2011; Knop and Kleinebudde, 2013; Razuc et al., 2019). Blend uniformity is one of the CQA that has been widely monitored using these spectroscopic methods (Corredor et al., 2014; De Beer et al., 2011; Nagy et al., 2017; Razuc et al., 2019).

The pharmaceutical powder processing units can cause significant changes in material properties (Escotet-Espinoza et al., 2018b). Thus, the monitoring of blend uniformity using a spectroscopic sensor mounted on the tablet press feed frame capable of detecting the effect of all previous processing units has become an emerging technique. The feed frame is the last place to access powder for in-line analysis just before it enters the dies for compression, this guarantees the representativeness of the measurements in blend uniformity, with respect to the drug concentration of the tablets (Sierra-Vega et al., 2019). An advantage of selecting the tablet press feed frame as sampling unit is that it is applicable to monitoring blend uniformity in both batch and continuous manufacturing processes (Esbensen et al., 2018; Sierra-Vega et al., 2019). Modern NIR spectrometers can produce spectra every

1–5 s for the blend flowing through the feed frame. The analysis at the feed frame does not depend on pre-selected sampling units as when sample thieves are used (Romañach, 2015). All parts of a lot have the same opportunity of being sampled when real-time analyses are performed at the feed frame. Spectroscopic monitoring can also be achieved at the feed frame without interrupting a pharmaceutical process or affecting the properties of the powder blend, another critical advantage over the use of sample thieves. However, the localization of the NIR probe must be carefully selected to guarantee reproducibility in the measurements and a non-stagnation of the powder (Martinez-Cartagena et al., 2019; Mateo-Ortiz et al., 2014a; Sierra-Vega et al., 2019). Fig. 7(a) and (b) show two feed frames with NIR probes for the acquisition of in-line spectra. The geometry and size of the probe play a significant role in the experimental set up (Sierra-Vega et al., 2019). Fig. 7(c) shows a feed frame set up when using simultaneously different PAT tools for in-line monitoring in the tablet press feed frame.

The first investigation regarding in-line monitoring in the tablet press feed frame evaluated the effect of probe-to-paddle distance and the paddle wheel speed on the dynamic sample presentation within the feed frame (Liu and Blackwood, 2012). The results showed that the paddle wheel speed generated oscillations in the NIR measurement with a frequency directly proportional to its speed. However, these oscillations tended to disappear when the probe-to-paddle distance increased. The NIR signal was also correlated with the tablet potency from stratified tablet samples at 3.5% w/w, while compression process parameters such as mass flow rate, quantity of placebo initially charged, and paddle wheel speed were varied (Ward et al., 2013). Excellent agreement was found between the NIR signal and off-line tablet potency measurements when the flow rate and initial placebo loading were varied. However, a high correlation between the paddle wheel speed and NIR signal was only found at higher paddle wheel speeds. Similarly, in-line NIR spectra were acquired to a formulation of 3.5% w/w API concentration within the feed frame, showing that the NIR signal at a single wavelength associated with the API has a high correlation with tablet potency (Šašić et al., 2015). However, the authors specify that this relationship is not satisfactorily transferable.

NIR spectroscopy was used to quantify API concentration (30% w/w) and two excipient concentrations (65% w/w in total). The NIR calibration models were developed from off-line data, transferring the model using Local Centering method transfer (Wahl et al., 2014). The in-line model was able to predict the concentrations of independent validation samples with root mean square error of prediction (RMSEP) less than 2.5% w/w. In a previous paper, a NIR calibration model was developed to quantify in real-time the drug concentration of powder blends through the feed frame using a simulated die disc (Mateo-Ortiz et al., 2014a). The effectiveness of the in-line over off-line NIR calibration models for 5–15% w/w API concentration blends in the feed frame was evaluated. The RMSEP for the in-line model was 50% lower than the RMSEP reported for the off-line model. Also, this study demonstrated that a 50% change in the paddle wheel speed and the distance between the NIR probe and powder bed affect the NIR spectral baseline and the NIR predictions. The same methodology was used in another study (Hetrick et al., 2017) where a NIR calibration model was developed to quantify drug concentration in real-time through the feed frame in a continuous manufacturing process, based on the spectral data acquired from feed frame with a simulated die disc. Variations in the mass flow, paddle wheel speed, and relative humidity were included in the NIR model. Excellent accuracy and precision results were obtained when the NIR model was applied to independent test runs, on both in the feed frame with a simulated die disc and in the continuous manufacturing line, demonstrating the equivalency between the off-line feed frame table and the full continuous manufacturing process.

Two in-line NIR calibration models were developed and validated to quantify the sodium saccharine concentration (5 and 20% w/w) in the feed frame of an industrial high speed rotary tablet press, Modul^{TM}P (GEA Pharma Systems – GEA Process Engineering, Halle, Belgium) (De Leersnyder et al., 2018). The effect of process parameters on the prediction of the 20% w/w NIR calibration model was evaluated, finding that the die disc speed does not affect the predictive performance of the model, and the low paddle wheel speed causes higher standard deviation in NIR predictions. The effect of paddle wheel speed and die disc speed on the NIR prediction to quantify 3% w/w acetaminophen was also studied (Sierra-Vega et al., 2018). The NIR calibration model showed a relative standard error of prediction (RSEP (%)) of 2.8%. The paddle wheel speed can be increased up to 30% without having a significant effect on the NIR predictions, while the die disc speed can only be decreased up to 10%. This last observation differs from that presented by Leersnyder et al. (De Leersnyder et al., 2018), possibly due to the differences between the concentration level used in each study, because at low concentration the physical effects stand out and small physical changes can affect the NIR predictions (Alcalá et al., 2008; Peroza Meza et al., 2006).

The drug concentration of powder blends within the feed frame has also been compared with the drug concentration of tablet, as well as in the chute before the tablet press (Dalvi et al., 2019; Pauli et al., 2019; Sierra-Vega et al., 2019). The results showed a good relationship between NIR predicted concentrations within the feed frame and tablets with off-line NIR, in-line NIR, and UV analysis. However, the statistical results demonstrated that the in-line drug concentration in the powder blends in the chute of a continuous direct compaction pharmaceutical manufacturing process is not representative of the drug concentration of tablets due to the mixing effect within the feed frame (Sierra-Vega et al., 2019).

A NIR calibration model was developed to monitor the blend uniformity in a direct compression process considering the paddle wheel speed and variation in the microcrystalline cellulose concentration as additional factors (Li et al., 2019). The samples at different speeds were removed without causing a significant change in the model predictions. However, when the spectra with excipient variations were removed, the models were unable to determine the test set content accurately. The ability of the NIR spectroscopy to detect changes in bulk density and porosity of the powder blends within the feed frame was also demonstrated (Ortega-Zúñiga et al., 2019). Three NIR calibration models were developed for the in-line determination of drug concentration, bulk density, and relative specific void volume (porosity) of 3% w/w acetaminophen powder blends. The bulk density NIR calibration model and porosity NIR calibration model were able to predict the validation blends with RSEP (%) of 0.57 and 0.44%, respectively. Near-infrared chemical imaging (NIR CI) alone and in combination with NIR spectroscopy was also used to in-line monitor the ascorbic acid composition within the feed frame (Dalvi et al., 2018). The combination of NIR and NIR CI demonstrated a better performance to predict the test set blends. However, this work highlights the ability of NIR CI to monitor the feed frame in a similar way to NIR spectroscopy.

Light-induced fluorescence spectroscopy, RGB color imaging, and NIR spectroscopy were simultaneously used for in-line monitoring of multivitamin blends within the feed frame of a Manesty Novapress 37-station rotary tablet press (Durão et al., 2017; Gosselin et al., 2017). The components studied in the multivitamin blends were selected for their ability to be detected by NIR, RGB, or LIFS probes. These three probes can monitor the composition in multivitamin blends of 6-components (Gosselin et al., 2017). LIFS quantified low concentrations of Beta-Carotene (< 1% w/w) and Riboflavin (< 2% w/w), while NIR spectroscopy quantified Ferrous Fumarate (2–8% w/w), Ginseng and Ascorbic Acid (5–15% w/w) in commercial multivitamin blends of 31-components. However, RGB showed to be less efficient (Durão et al., 2017).

Raman spectroscopy has also been used to monitor API concentration within the feed frame (Li et al., 2018). This study evaluated the off-line, static, and in-line strategies to develop the principal component regression (PCR) models. The off-line model was performed in a spinning cup, and the static model was developed using static powders

Table 1
Summary of the spectroscopy methods used for in-line monitoring in the feed frame.

Method*	Monitored information	API Concentration range or API target concentration (%w/w)	Evaluated challenges	Data-analysis method**	Calibration performance ***	[Reference] (Year)
NIRS	NIR absorbance and tablet potency	4%	Effect of movement of powder and the probe distance on the quality of spectra. Compare potency profile in the feed frame and in the stratified tablets.	PCA	–	(Liu and Blackwood, 2012)
NIRS	NIR absorbance at 1620 and 1669 nm and tablet potency	3.5–30%	Evaluate the correlation between the NIR signal and the tablet potency from stratified tablets samples.	–	–	(Ward et al., 2013)
NIRS	API concentration, excipients concentration	30%	Simultaneously quantify the concentration of an API and two main excipients in an industrial tablet press.	PCA, PLS	RMSECV: 4.4 RMSEP: 2.7 $R^2_{cal} = 0.97$	(Wahl et al., 2014)
NIRS	Acetaminophen concentration	5–15%	Use off-line and in-line approaches to develop NIR calibration model and evaluate the effect the 50% changes in the paddle wheel speed on NIR predictions.	PCA, PLS	RMSEC (Off-line): 1.07 RMSECV (Off-line):1.10 RMSEP (Off-line):0.80 Bias (Off-line): 0.03 RMSEC (in-line): 0.37 RMSECV (in-line):0.38 RMSEP (in-line): 0.40 Bias (in-line): −0.11	(Mateo-Ortiz et al., 2014a, b)
NIRS	API concentration	3.5%	Monitoring the API concentration in powder blends using univariate analysis and second derivative as data pretreatment. PCA also was used as additional source of information.	PCA	–	(Šašić et al., 2015)
NIRS, LIFS, RGB color imaging	Vitamin concentration	Vitamin I: 5–11% Vitamin II: 0–4% Vitamin III: 0–2%	Monitoring the concentration of three vitamins in powder blends of six-components using three PAT tools alone and combined.	PCA, MIA	–	(Gosselin et al., 2017)
NIRS, LIFS, RGB color imaging	Vitamin concentration	Beta-Carotene: < 1% Riboflavin: < 2% Ferrous Fumarate: 2–8% Ginseng: 2–10% Ascorbic Acid: 5–15% Pyridoxine > 1.5%	Monitoring the concentration of five key components of a commercial multivitamin blends of 31-components using three PAT tools alone and combined.	PCA, PLS, MIA	R^2_{cal} greater than 0.90	(Durão et al., 2017)
NIRS	API concentration	Not specified	Quantify API concentration of powder blends in a continuous manufacturing process (CM), based on the spectral data acquired from a “feed frame table”.	PLS	RMSEC (feed frame table): 2.14 RMSECV (feed frame table):2.14 RMSEP (feed frame table)≅ 0.75 RMSEP (CM): 1.3	(Hetrick et al., 2017)
NIRS	Sodium Saccharine concentration	Range 1: 3–7% Range 2: 15–25%	Monitoring the API concentration (5 and 20% w/w) and evaluate effect of paddle design, paddle wheel speed and tableting speed on predictive performance.	PLS	RMSEP (Model 5%): 0.105 to 0.784 RMSEP (Model 20%): 0.791 to 1.734	(De Leersnyder et al., 2018)
NIRS, NIR CI	Ascorbic acid	0–12%	Use NIR CI alone and in combination with NIRS to in-line monitor the ascorbic acid composition.	PCA, PLS	$RMSECV_{NIR}$: 0.71 $RMSEP_{NIR}$: 0.72 to 2.53 $R^2_{cal}(NIR) = 0.96$ $RMSECV_{NIR\ CI}$: 0.82 $RMSEP_{NIR\ CI}$: 0.98 to 2.02 $R^2_{cal}(NIRCI) = 0.95$ $RMSECV_{NIR/NIR\ CI}$: 0.51 $RMSEP_{NIR/NIR\ CI}$: 0.53 – 2.18 $R^2_{cal}(NIR/NIRCI) = 0.98$	(Dalvi et al., 2018)
NIRS	Acetaminophen concentration	1.5–4.5%	Quantify the API concentration (3% w/w) and evaluate the powder flow behavior at different paddle wheel and die disc speeds.	PCA, PLS	RMSEP: 0.083 RSEP: 2.778 Bias: 0.004	(Sierra-Vega et al., 2018)

(*continued on next page*)

Table 1 (*continued*)

Method*	Monitored information	API Concentration range or API target concentration (%w/w)	Evaluated challenges	Data-analysis method**	Calibration performance ***	[Reference] (Year)
Raman	Acetaminophen concentration	8%	Use the off-line and in-line approaches to develop a calibration model to quantify API concentration. Evaluate the model robustness by changing the press speed.	PCA, PCR	RMSECV (in-line):1.75 RMSEP (in-line): 1.30 to 2.03 RMSECV (static):3.06 RMSEP (static): 1.54 to 4.65 RMSECV (off-line):2.69 RMSEP (off-line): 11.25 to 12.60	(Li et al., 2018)
NIRS	Acetaminophen concentration, blend bulk density and blend relative specific void volume	1.5–4.5%	Use the physical and chemical information present in the NIR spectra to develop NIR calibration models for the determination of drug concentration, bulk density and porosity.	PCA, PLS	RMSEP (concentration): 0.07 to 1.25 RSEP (concentration): 2.21 to 41.51 RMSEP (density): < 0.04 RSEP (density):0.44 to 4.45 RMSEP (porosity): 0.01 to 0.25 RSEP (porosity): 0.57 to 7.76	(Ortega-Zúñiga et al., 2019)
NIRS	Ibuprofen concentration	0–15.3%	In-line monitoring the ibuprofen concentration in powder blend and compressed tablets.	PLS	RMSEC (powder):0.46 RMSECV (powder): 0.47 $R^2_{cal}(powder) = 0.995$ RMSEC (tablets):1.83 RMSECV (tablets): 1.87 $R^2_{cal}(tablets) = 0.92$	(Dalvi et al., 2019)
NIRS	Acetaminophen concentration	10%	Monitoring blend uniformity throughout a continuous manufacturing process by near infrared (NIR) spectroscopic measurements of flowing blends and compared to the drug concentration of powder blends in the chute and drug concentration of tablets.	PCA, PLS	RMSEP: 0.16 RSEP: 1.58 Bias: 0.05 $R^2_{cal} = 0.99$	(Sierra-Vega et al., 2019)
NIR	Diclofenac Sodium	25%	Monitoring blend uniformity of dried granulates in the feed frame and compared to tablet content uniformity. Changes in tableting speed up to 70,000 tablets/h were considered.	PCA, PLS	RMSEC (powder):1.8 RMSEP (powder): 1.8 $R^2_{cal}(powder) = 0.99$ RMSEC (tablets):2.2 RMSEP (tablets): 2.3 $R^2_{cal}(tablets) = 0.99$	(Pauli et al., 2019)
NIR	Acetaminophen concentration	6–10%	Monitoring blend uniformity in a direct compression process and study the effect of paddle speed and the variations in the excipient's concentration on method robustness.	PCR	RMSEC: 2.64 RMSEP: 2.85 $R^2_{cal} = 0.99$	(Li et al., 2019)

*NIRS: Near infrared spectroscopy, LIFS: Light-induced fluorescence spectroscopy, RGB: Red, green and blue color imaging.

** PCA: Principal Component Analysis, PCR: Principal Component Regression, PLS: Partial Least Square, MIA: Multivariate Image Analysis.

*** RMSEC: root mean square error of calibration (%w/w); RMSEP: root mean square error of prediction (%w/w); RMSECV: root mean square error of cross-validation (%w/w); RSEP: relative standard error of prediction (%); R^2_{cal}: correlation coefficient of calibration; R^2_{val}: correlation coefficient of validation.

within the feed frame. The in-line PCR model provided better predictions of the independent test set than static and off-line models. However, the off-line model was able to predict the API concentration after bias correction. The robustness of the PCR model was evaluated with a 10% variation in the tableting speed, demonstrating that these variations did not affect the PCR model predictions.

The studies analyzed in this section show that the tablet press feed frame is an excellent option for in-line monitoring of blend uniformity, with a precision and accuracy equal to or greater than traditional off-line methods, but with a significantly lower analysis time (De Beer et al., 2011; Vargas et al., 2018). In fact, one of the four commercial products approved for continuous manufacturing by the FDA (DiProspero, 2019; Pharmaceutical Technology, 2018, 2016) uses the feed frame as a sampling location for blend uniformity (Diller and Jerridge, 2009; Hetrick et al., 2017). Table 1 shows a summary of the studies presented in which spectroscopic techniques were used for in-line monitoring of powder blends within the feed frame. For each study: the method, monitored information, API concentration range or target concentration, evaluated challenges, and calibration performance are specified.

5.1. Effective sample size within the feed frame

The FDA draft guidance on the development and submission of NIR analytical procedures suggests that for blend uniformity analysis, the effective sample size should be comparable to a unit dose (FDA, 2015). The effective sample size estimation depends on the sampling location for blend uniformity and the optical parameters of the probe (Bakri et al., 2015). Eq. (6) was proposed to estimate the effective sample size in a continuous blending process (Colón et al., 2014). This equation, derived for a sample that moves linearly and perpendicularly to the NIR probe, indicates that the mass analyzed depends directly on the bulk density of the powder blend (ρ), the spectroscopic probe size (d), the displacement of the powder blends during the analysis (Δx) and the depth of penetration of the light (H). This equation also assumes that the powder area analyzed is rectangular. The displacement of the powder is calculated based on the acquisition time of one spectrum, and the linear velocity of the powder blends.

$$M = \rho\left[\pi\left(\frac{d}{2}\right)^2 + d(\Delta x)\right]H \tag{6}$$

The powder flow dynamics within the feed frame differs greatly from this behavior. Therefore, Eq. (6) was modified to obtain Eq. (7) to estimate the effective sample size within the feed frame (Li et al., 2018). In addition of bulk density (ρ) and fixed depth of penetration (H), Eq. (7) considers the angular paddle wheel speed (ω), the beam radius (r), the distance between the center of the probe and the center of the paddle wheel (l), the time to acquire a scan of the spectrum (t), and assumes the geometric shape of the analyzed volume by the probe as a portion of a hemi-ellipsoid.

$$M = \rho\pi rH\left[\frac{t}{2}\omega l + \frac{2}{3}r\right] \tag{7}$$

Table 2 summarizes the studies that have reported the effective sample size within the feed frame as a function of beam diameter, integration time, number of scans, paddle wheel speed and depth of penetration. An increase in paddle wheel speed, in integration time and the number of scans, causes an increase in the effective sample size (Li et al., 2019). These optical parameters must be carefully evaluated before developing a PAT method in the feed frame.

According to Eq. (6) and (7), the most significant sources of error to estimate the effective sample size are the fixed depth of penetration and the bulk density in the process. The depth of penetration depends on the wavelength, physical properties (particle size, the density, and cohesion), the chemical composition of the powder blend, and the optical system (Huang et al., 2016; Li et al., 2019; Razuc et al., 2019; Shi and Anderson, 2010). The depth of penetration has been estimated statically by different means (Bakri et al., 2015; Berntsson et al., 1999; Huang et al., 2016; Ortega-Zúñiga et al., 2019; Román-Ospino et al., 2018). However, the depth of penetration in the feed frame depends on powder wave behavior and the mass hold-up. Different distances between the probe and the powder bed affect the NIR predictions and this may be a consequence of different penetration depths (Colón et al., 2017; Mateo-Ortiz et al., 2014a).

The density within the feed frame is not easily estimated, as the bulk density within the feed frame depends on the consolidation of the powder blends and changes due to paddle wheel speed (Dalvi et al., 2019; Mateo-Ortiz et al., 2014a). The bulk density measured off-line for the estimation of the effective sample size by Eqs. (6) or (7) does not consider the powder dynamic within the feed frame. The powder blend within the feed frame has different densities, these densities do not reach the degree of consolidation to be considered a tap density, but they are higher than the bulk density (Mateo-Ortiz and Méndez, 2016). Therefore, an average between the bulk density and the tap density is recommended for the estimation of the effective sample size when monitoring the powder blends within the feed frame.

5.2. Variographic analysis in the feed frame

The analytical error is usually reported in all the methods developed in the pharmaceutical industry, but the sampling error is not commonly reported. Sampling procedures are often focused on the number of samples acquired and not on the way in which they are acquired (Esbensen et al., 2016; Esbensen and Paasch-Mortensen, 2010). Variographic analysis has been reported as a suitable method to estimate the magnitude of the total error measurements (sampling and analytical error) (Esbensen et al., 2018, 2016; Sánchez-Paternina et al., 2019). Variographic analysis makes it possible to discriminate between the process variation and the sampling and analytical errors (Esbensen et al., 2007). This technique has been used extensively for more than 60 years in industries such as mining, metallurgical, and petroleum (Esbensen et al., 2016). However, it was only recently applied to the pharmaceutical industry (Colón et al., 2017, 2014; Sánchez-Paternina et al., 2019; Sierra-Vega et al., 2019, 2018; Vargas et al., 2018, 2017).

Variographic analysis leads to the calculation of the variogram (Esbensen et al., 2007; Esbensen and Julius, 2010). The variogram is calculated employing the variogram function, $V(j)$, described in Eq. (8), where $V(j)$ represents the variance between extracted increments. Q_{total}

Table 2
Effective sample size reported for blend uniformity analysis within the feed frame.

Study	Method	Spot size (mm)	Integration time/acquisition time (ms)	Number of scans	Paddle wheel speed (rpm)	Depth of penetration (mm)	Estimated mass	Reference
1	NIRS	6.0	50	150	25	0.5	15 mg/cm^2	(Wahl et al., 2014)
2	NIRS	6.0	–	15	–	1.0	435 mg	(De Leersnyder et al., 2018)
3	RAMAN	3.0	3000	2	15	0.5	1130 mg	(Li et al., 2018)
4	NIR	5.0	8	64	15	0.5	90 mg	(Li et al., 2019)
5	NIRS	–	5	50	13	1.5	299.6 mg	(Dalvi et al., 2019)

represents the total number of analytical results (predicted concentrations), h is the heterogeneity contribution by each sample (concentration value), and j is the lag or distance between pairs of samples. A lag of one is the unit distance (time or distance) between two consecutive concentrations predicted by the analytical method. The order in which the samples are acquired is strictly necessary for the calculation of the variogram.

$$V(j) = \frac{1}{2(Q_{total} - j)} \sum_{q=1}^{Q_{total}-j} (h_{(q+j)} - h_q)^2 \quad (8)$$

In variographic analysis, three parameters are calculated: nugget effect, range, and sill. These terms originate from the use of the variogram in the geochemical industry (Gy, 1998). The nugget effect (Minimum possible error, MPE) corresponds to the sum of the total analytical error (TAE) and total sampling error (TSE) related with the measurement system used (Esbensen et al., 2007). The MPE is calculated by extrapolating the function *V(j)* to the *Y-axis* (lag = 0). A "lag 0" is not physically possible but provides an estimate of the minimum sampling errors under specific conditions. The sill delivers information about the total process variation, and the range is the lag where the sill is achieved. Fig. 8(a) and (b) show two different variograms, in both variograms the MPE, the sill and range have been identified. Fig. 8(a) shows a variogram with a significantly high sill value with respect to the MPE. This indicates a high heterogeneity where improvements are needed. Fig. 8(b) shows a variogram with a sill value close to the MPE. This represents a process with low process variation and low sampling and analytical errors.

The first variogram for a feed frame was published as recently as 2018 (Sierra-Vega et al., 2018). Variograms with significantly low MPE (MPE < 0.0035 $(\%w/w)^2$) and sill values very close to the MPE (sill < 0.006 $(\%w/w)^2$) were found, indicating low sampling and analytical errors and low total process variation in the feed frame. Variographic analysis was also used to compare the performance of a chute after the continuous blender and the tablet press feed frame to monitor powder blend uniformity in a continuous direct compaction manufacturing process (Sierra-Vega et al., 2019). This study demonstrated that the feed frame reduces the process variation present in the chute, showing a sill six times less than the chute. Also, this study demonstrated that the feed frame presents an MPE approximately five times less than the chute. These studies improve the understanding of the sources of error in the analytical methods developed in the feed frame in agreement with the objectives of the FDA process validation guidance (FDA, 2011).

6. Conclusions and future challenges

The studies that provide valuable information on powder dynamics within the feed frame were summarized, demonstrating the importance of this process unit in the pharmaceutical tablet manufacturing. This review shows that the paddle wheel speed and die disc speed must be established based on the blend properties to obtain an adequate weight and minimize weight variability and undesired effects, such as particle attrition and over-lubrication that affect the CQA of the tablets. This review also shows that the feed frame is an excellent option to monitor blend uniformity, showing a high agreement with the drug concentration of the tablets. NIR spectroscopy has been the most implemented analytical technique to monitor drug concentration within the feed frame, quantifying drug concentration as low as 3% w/w with a low analytical error.

The feed frame is a unit with evident importance in recent years, and based on present review, it is possible to identify opportunities of studies in this process unit. Simulations are a valuable tool to understand the powder flow behavior within the feed frame and to design or evaluate new paddle designs. However, simulations in this field should focus on working with reduced particle sizes and developing its ability to handle increased cohesion, which requires more computing power. The adjustment of these parameters leads to having a more accurate representation of the powder flow dynamics within the feed frame.

Efforts focusing on the connection between the RTD profile and the mixing effect within the feed frame are required. The mixing effect has been addressed in only a few publications and is fundamental for the understanding of the powder flow behavior within the feed frame, its impact on blend uniformity and its effect on the tablet properties. More studies are needed to understand the mixing effect as a function of the powder blend properties, the powder blend heterogeneity in the input of the feed frame, the operating conditions, and paddle wheel design. Likewise, research that focuses on the influence of the particle size segregation, weight adjustment, the suction effect of the tablet press on the RTD and its parameters are necessary.

Although in-line monitoring in the feed frame has been successful, more considerable efforts are required for the feed frame to be accepted and implemented as a sampling unit within the control strategy in pharmaceutical tablet manufacturing. Additional researches are needed to study the use of granulated materials to develop in-line models. This will allow studying the effect of particle attrition within the feed frame on the performance of the calibration model. Similarly, studies that investigate the effect of cohesion on the performance of the calibration modes are needed, since powder flow behavior is significantly affected by cohesion, which can be an additional challenge when developing in-line calibration models.

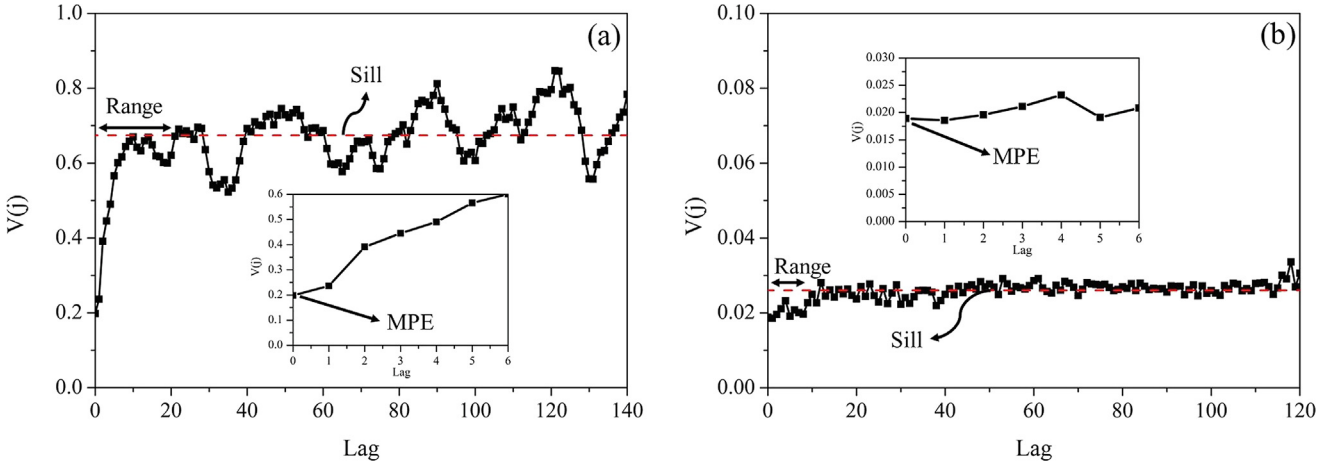


Fig. 8. Variographic Analysis: Identification of nugget effect, range, and sill.

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